The Brain and Nervous System

How a signal travels, which chemicals carry it, what each part of the brain does, and how to spot the neuroscience claims that sell well but don't hold.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 2 — rewritten, updated and applied

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Every behaviour in this series — a purchase, a memory, a flash of anger in a review meeting — is in the end something the nervous system did. You don't need to be a neurologist to manage people. But two things make this part worth the effort. First, it explains the body's stress response, which every target-driven job switches on daily. Second, "neuro" is the most over-sold prefix in business: neuromarketing, neuro-leadership, brain-based training. Knowing the basics is how you tell the real from the pitch.

1 The neuron: the basic unit

The nervous system is built from **neurons**, cells specialised to carry messages. The human brain has roughly 86 billion of them. Each has three working parts:

- **Dendrites** — branches that receive signals from other neurons.
- **Cell body (soma)** — keeps the cell alive and adds up incoming signals.
- **Axon** — a long fibre that carries the signal away to the next cell. Many axons are wrapped in **myelin**, a fatty insulation that makes signals travel much faster.

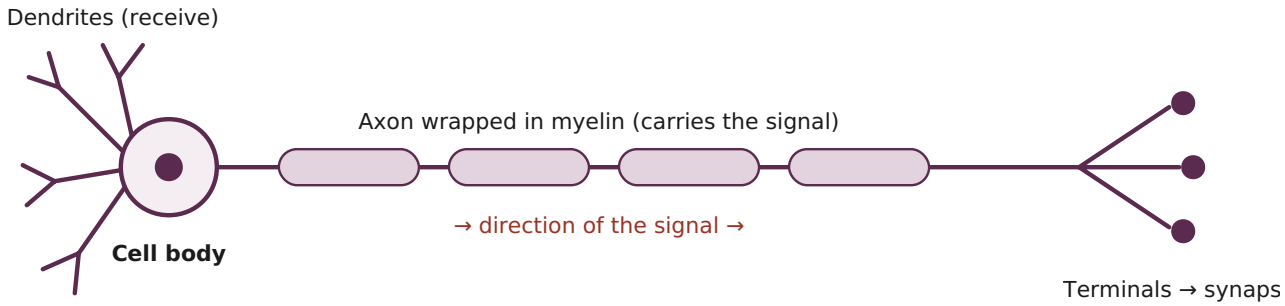


Figure 1. A neuron. Signals come in through the dendrites, are summed in the cell body, and travel down the axon to the terminals.

A neuron either fires or doesn't — the **all-or-none** principle. It fires when the inputs it receives push it past a threshold; the electrical pulse that then runs down the axon is the **action potential**. A strong stimulus doesn't make a bigger pulse. It makes the neuron fire more often, and recruits more neurons.

Neurons don't touch. At the end of the axon is a tiny gap, the **synapse**. The electrical signal is converted into chemical messengers, **neurotransmitters**, which cross the gap and either excite the next neuron (make it more likely to fire) or inhibit it (less likely). Most psychiatric drugs work at the synapse, by changing how much of a neurotransmitter is released or how long it lingers.

Neurotransmitter	Main roles	Worth knowing
Glutamate	The main excitatory messenger; learning and memory	Involved in nearly every fast signal in the brain.
GABA	The main inhibitory messenger; calms activity	Anti-anxiety drugs and alcohol boost its effect.
Acetylcholine	Muscle movement; attention and memory	Lost early in Alzheimer's disease.
Dopamine	Movement; reward learning, motivation	Better described as "wanting" and prediction than as "pleasure" (see section 8).
Serotonin	Mood, sleep, appetite	Target of the most common antidepressants (SSRIs).
Norepinephrine	Alertness, arousal	Part of the stress response.
Endorphins	The body's own pain relievers	Released under stress, pain and hard exercise.

WORKED EXAMPLE 1 · WHY MYELIN MATTERS

A signal travels from your toe to your spinal cord, about 1.2 m. A well-myelinated axon carries it at about 100 m/s; an unmyelinated one at about 1 m/s.

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Myelinated:    time = 1.2 m / 100 m/s = 0.012 s = 12 ms
Unmyelinated:  time = 1.2 m / 1 m/s   = 1.2 s
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A hundredfold difference. Fast pain (the sharp "ouch") travels on myelinated fibres; the dull ache that follows comes on slower, unmyelinated ones. Multiple sclerosis is the immune system attacking myelin, which is why it slows and scrambles movement and sensation. Note the scale too: a typical human reaction time is around 200–250 ms, so most of the delay is the brain processing the signal, not the wire carrying it.

2 The map of the nervous system

The system has two halves. The **central nervous system (CNS)** is the brain and spinal cord — the decision-maker. The **peripheral nervous system (PNS)** is everything else — the wiring that connects the CNS to the body.

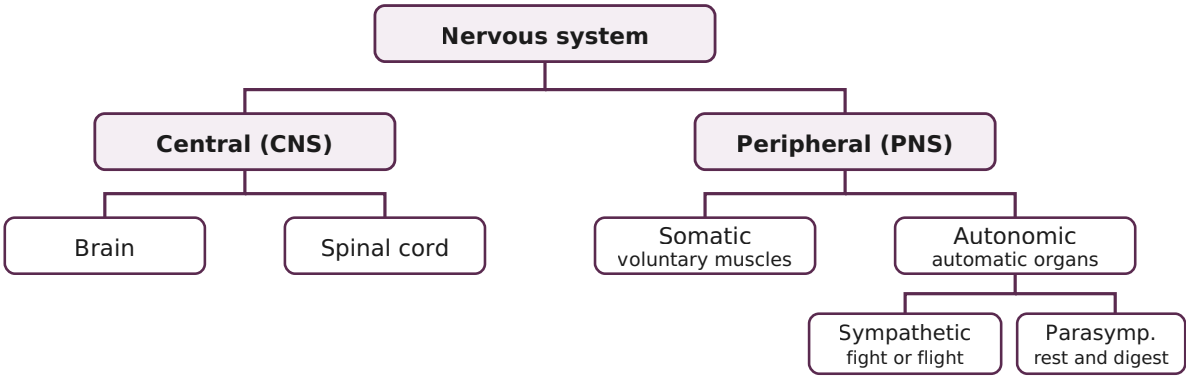


Figure 2. The divisions of the nervous system.

- The **spinal cord** carries messages up and down, and also handles simple **reflexes** on its own — you pull your hand off a hot pan before the brain registers pain.
- The **somatic** nervous system controls voluntary muscles and carries sensation in.
- The **autonomic** nervous system runs the organs without your say-so. Its **sympathetic** branch prepares for emergencies: heart rate and blood pressure up, digestion paused, pupils wide, sugar released for energy. Its **parasympathetic** branch reverses all that once the threat passes.

The sympathetic system cannot tell a leopard from a quarterly review. A public dressing-down in a sales meeting triggers the same racing heart and dry mouth as physical danger. That response was built for short, physical emergencies. Part 9 covers what happens when it stays switched on for months.

3 The endocrine system: slower messengers

Alongside the nerves runs a second communication system: **glands** that release **hormones** into the blood. Hormones are slower than nerve signals — seconds to hours rather than milliseconds — but their effects last longer.

- The **pituitary gland**, controlled by the brain's hypothalamus, is called the master gland because it directs the others.
- The **adrenal glands** release **adrenaline** (fast, during the fight-or-flight response) and **cortisol** (slower; mobilises energy during prolonged stress, and in excess over long periods it harms health).
- **Oxytocin** is involved in childbirth, breastfeeding and bonding. **Testosterone** and **oestrogen** are the main sex hormones. Both oxytocin and testosterone feature in popular business claims that the evidence does not support well (section 8).

4 How we look inside

Much of what we know came from damage: studying what a person loses after a stroke or injury to one area. Modern tools look at living, healthy brains:

- **EEG** records electrical activity through the scalp. Excellent timing (milliseconds), poor location. Used for sleep research (Part 3).
- **CT** and **MRI** scans show *structure* — a detailed picture of the brain, but not what it's doing.
- **fMRI** shows *activity* indirectly, by tracking oxygen-rich blood flowing to busy regions. Good location, but the blood response lags the neural activity by several seconds.
- **PET** tracks a mildly radioactive tracer, showing which areas use the most energy.

Keep the fMRI limitation in mind: a brightly coloured scan shows where blood flow rose by a few per cent, averaged across people and many trials. It is not a picture of a thought.

5 The brain from the bottom up

A useful way to picture the brain is in layers, from the oldest structures that keep you alive to the newest that let you plan a quarter.

The brainstem and cerebellum: life support and coordination

The **medulla** controls breathing and heart rate. The **pons** relays signals and helps regulate sleep. The **reticular formation**, a network running through the brainstem, controls alertness and filters which incoming information gets attention. The **cerebellum** at the back coordinates movement and balance and stores learned motor skills — typing, riding a bike, driving.

The limbic system: emotion and memory

- **Thalamus** — the relay station; nearly all incoming senses (except smell) pass through it on the way to the cortex.
- **Hypothalamus** — small but powerful: hunger, thirst, body temperature, sex drive, and control of the pituitary. It links the brain to the hormone system.
- **Hippocampus** — essential for forming new long-term memories (Part 5).
- **Amygdala** — involved in fear and in flagging emotionally important events, which is why emotional events are remembered better.

The cortex: the outer layer

The **cerebral cortex** is the wrinkled outer layer, folded so that a large surface fits inside the skull. It is divided into four lobes on each side:

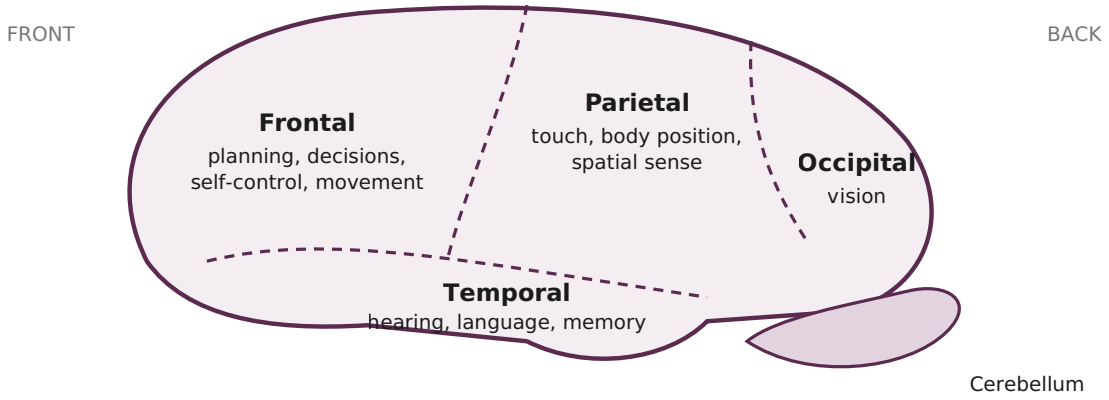


Figure 3. The four lobes of the cortex, seen from the left side. A schematic, not an anatomical drawing.

- **Occipital lobe** (back) — processes vision.
- **Parietal lobe** (top) — touch, temperature, body position; its front strip, the **somatosensory cortex**, maps the body's surface.
- **Temporal lobe** (sides) — hearing and understanding language (**Wernicke's area**, usually on the left).
- **Frontal lobe** (front) — planning, judgement, self-control, personality, and producing speech (**Broca's area**). Its rear strip, the **motor cortex**, controls voluntary movement.

Most of the cortex is **association areas**, which combine information rather than handle one sense. The frontal association areas, the **prefrontal cortex**, are the last to mature — into the mid-20s — and are the most sensitive to fatigue, alcohol and stress.

The classic case is **Phineas Gage**, a railway foreman who in 1848 survived an iron rod driven through his frontal lobe. Accounts described a sober, capable man becoming impulsive and unreliable. Historians now think the later stories were exaggerated and that Gage recovered more than the textbooks once said. But the core lesson has held: the frontal lobe is central to planning and self-control.

6 Two hemispheres

The brain has a left and a right half, joined by a thick band of fibres, the **corpus callosum**. Each half mostly controls the opposite side of the body. In most people language is handled mainly on the left; the right is relatively stronger at some spatial and face-recognition tasks.

The evidence comes from **split-brain** patients (Roger Sperry and Michael Gazzaniga, from the 1960s), whose corpus callosum had been cut to control severe epilepsy. Shown a word only to the right hemisphere, they couldn't say it, but could pick out the object with the left hand. One striking finding: the speaking left hemisphere would confidently invent a reason for actions it hadn't caused. Gazzaniga called it the **interpreter** — and it is a useful warning about how readily people explain their own behaviour after the fact.

7 Plasticity: the brain changes with use

The brain is not fixed after childhood. **Neuroplasticity** is its ability to rewire with experience: strengthening connections that are used and pruning ones that aren't. London taxi drivers, who must memorise thousands of streets, were found to have a larger rear hippocampus than comparison groups, and the size tracked years on the job. Skills built by deliberate practice are, physically, changes in the brain. That is the biological basis for the claim that capability is trainable, not fixed.

8 Held up? Five neuro-claims checked

Claim	Verdict	Why
"We only use 10% of our brain"	**False**	Imaging shows activity across the whole brain over a day; damage almost anywhere causes some deficit.
"People are left-brained (logical) or right-brained (creative)"	**False**	Specific functions do lean to one side, but a 2013 University of Utah scan study of over 1,000 people found no evidence that individuals are left- or right-brain dominant overall.
"Dopamine is the pleasure chemical"	**Oversimplified**	Dopamine tracks wanting and *prediction* — it spikes for an unexpected reward and dips when an expected one fails to come. That is closer to "craving" than "enjoyment".
"A sniff of oxytocin makes people trust you"	**Not supported**	A famous 2005 study found nasal oxytocin raised trust in an investment game. Later replications and reviews found the effect unreliable.
"Brain scans show what consumers really want"	**Mostly overstated**	Scans can add some predictive power, but the usual claim relies on reverse inference (worked example 2).

WORKED EXAMPLE 2 · THE REVERSE-INFERENCE TRAP

A neuromarketing agency shows that a client's ad "lights up the amygdala" and claims viewers felt fear-driven urgency. Suppose (illustrative numbers):

- When people genuinely feel fear, the amygdala activates 80% of the time.
- But it also activates for 30% of non-fear moments — anything novel, surprising or important.
- While watching ads, people feel genuine fear only 10% of the time.

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P(fear | amygdala active)
  = P(active | fear) × P(fear) / P(active)
  = (0.8 × 0.1) / (0.8 × 0.1 + 0.3 × 0.9)
  = 0.08 / (0.08 + 0.27) = 0.08 / 0.35 ≈ 0.23
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Given activation, the chance the viewer felt fear is only about 23%. Most brain regions do many jobs, so "region X lit up, therefore the person felt Y" is weak unless the region is highly specific to Y. Ask any neuro-vendor: how specific is that region?

Summary

- **Neurons** receive signals through dendrites and send them along the axon; **myelin** speeds them up. Firing is **all-or-none**; intensity is coded by how often neurons fire.
- At the **synapse**, **neurotransmitters** carry the signal chemically. Glutamate excites, GABA calms; dopamine is about wanting and prediction more than pleasure; serotonin relates to mood.
- The **CNS** (brain and spinal cord) decides; the **PNS** connects. Its **sympathetic** branch triggers fight-or-flight and the **parasympathetic** calms the body. Workplace threats trigger the same response as physical danger.
- **Hormones** are slower but longer-lasting. Adrenaline and cortisol drive the stress response.
- EEG has good timing, fMRI good location; neither is a picture of a thought.
- Brainstem: life support. Cerebellum: coordination and motor skills. Limbic system: emotion and memory (amygdala, hippocampus). Cortex: four lobes, with the **prefrontal cortex** handling planning and self-control.
- The hemispheres specialise in some functions, but "left-brained" and "right-brained" people don't exist.
- **Neuroplasticity**: the brain rewires with practice, so skills are physically trainable.
- Treat neuromarketing claims with care: reverse inference is weak unless the brain region is specific.

Practice problems

P1. A signal must travel 0.9 m along an axon. How long does it take at 80 m/s, and at 2 m/s?

P2. An ASM is told in a review meeting that his territory will be split. His heart races and his mouth goes dry. Which division of the nervous system is responsible? Name two other changes it causes.

P3. Match each to a lobe: (a) recognising a customer's face across the store, (b) understanding what a dealer is saying on the phone, (c) deciding not to snap back at a rude dealer, (d) feeling the weight of a paint tin in your hand.

P4. A team-building vendor says its workshop "rebalances left-brain logic with right-brain creativity". What would you tell your HR head?

P5. Redo worked example 2 with a more specific region: it activates 80% of the time during the emotion and only 5% of the time otherwise; the base rate is still 10%. What is P(emotion | active)?

P6. Why might a salesperson who drove 400 km and slept five hours make a worse pricing decision that evening? Use this part's vocabulary.

Glossary

Neuron

A cell that carries information through the nervous system.

Dendrites / axon

The branches that receive signals / the fibre that carries them away.

Myelin

Fatty insulation around axons that speeds up signals.

Action potential

The electrical pulse that runs down the axon when a neuron fires.

All-or-none

A neuron fires fully or not at all.

Synapse

The gap between neurons, crossed by neurotransmitters.

Neurotransmitter

A chemical messenger that excites or inhibits the next neuron.

Sympathetic / parasympathetic

Branches of the autonomic system that arouse the body / calm it.

Hormone

A chemical released by a gland into the blood.

Cortisol

A stress hormone that mobilises energy during prolonged stress.

fMRI

A scan that tracks blood flow to show which areas are active.

Amygdala / hippocampus

Limbic structures for emotional significance / forming new memories.

Prefrontal cortex

Front of the frontal lobe; planning, judgement, self-control.

Corpus callosum

The band of fibres joining the two hemispheres.

Neuroplasticity

The brain's ability to change its connections with experience.

Reverse inference

Concluding a mental state from brain activity in a region that serves many functions.

Answer key

P1. 0.9 / 80 = 0.011 s ≈ 11 ms. 0.9 / 2 = 0.45 s.

P2. The sympathetic branch of the autonomic nervous system. Also: higher blood pressure, faster breathing, wider pupils, digestion slowed, sugar released into the blood, adrenaline from the adrenal glands.

P3. (a) occipital, with temporal areas handling face recognition; (b) temporal (Wernicke's area); (c) frontal (prefrontal cortex); (d) parietal (somatosensory cortex).

P4. Functions lean to one side, but people are not left- or right-brain dominant; a large 2013 scan study found no such personality split. The claim is a myth, so judge the workshop on any other evidence it has.

P5. (0.8 × 0.1) / (0.8 × 0.1 + 0.05 × 0.9) = 0.08 / 0.125 = **0.64**. Specificity raised the confidence from 23% to 64%.

P6. Fatigue and sleep loss impair the prefrontal cortex first, weakening planning and self-control. Long-running stress keeps the sympathetic system and cortisol elevated. Together these favour quick, impulsive choices — such as conceding a discount to close.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 2; text and figures are original. Left/right brain: Nielsen et al., *PLOS ONE*, 2013. Taxi drivers: Maguire et al., *PNAS*, 2000. Oxytocin and trust: Kosfeld et al., *Nature*, 2005, and later reviews of replication failures. Reverse inference: Poldrack, *Trends in Cognitive Sciences*, 2006. Worked-example numbers are illustrative.